Spatial dimension compactification in general relativity and duality symmetry

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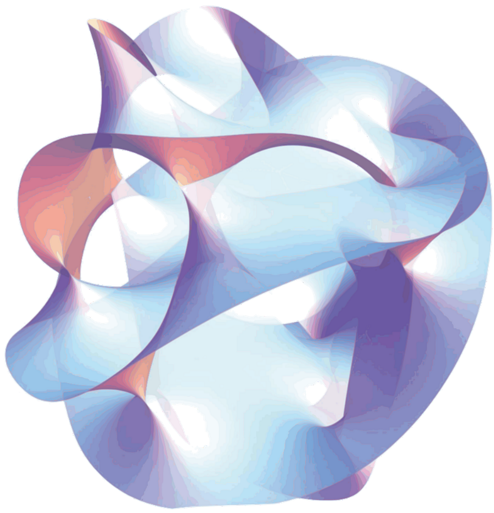

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Notations and abreviations

For the abreviations we wil use

- GR for general relativity

We will during this intership use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices $(\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots)$
- Latin hatted letter for flat indices $(\hat{p}, \hat{m}, \hat{n}, \dots)$
- For space-time indices taking D values, late greek letters are used in the curved case $(\mu, \nu, \rho, \dots)$ and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case $(\alpha, \beta, \gamma, \dots)$ and Latin letters in the flat case (a,b,c)

THe general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate x^{μ} and internal coordinate y^{α}. The flat lorenz

Contents

1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything else.

One of the earliest attempts to change this was proposed by Kaluza in 1921 [3] and later refined by Klein:
by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had problems, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [4] and string theory.

2 General relativity theoretical reminder

Here you will find some GR reminder, which hopefully will make it easier to understand the mathematical foundation on which this internship was build

2.1 mathematical basis

Definition 1 *Let V be an R-vector space of dimension N, we define the tensor of order* $(r, s) \in [\![0, 1]\!]^2$ *on* $V^{(r,s)} \overset{def}{=} V^{\otimes r} \otimes V^{\otimes s}$ *as a being a multilinear application on* $V^{\star r} \times V^s$ *to* $\mathbb{R}$

Definition 2 *We will call n-dimension* ($n \in \mathbb{N}^*$) *manifold every Hausdorff topological space* $(\mathcal{M}, \tau)$ *localy homeomorphic to* $\mathbb{R}^n$)

Definition 3 *Let* $(\mathcal{M}, g)$ *be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form* $g_{\mu\nu}$ *on the tangent space* $T_p\mathcal{M}$ *at each point* $p \in \mathcal{M}$*, such that*

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

In General Relativity we use the signature $(-, +, +, +)$.

Definition 4 *In general in RG we use the christofells connexion as it is the sole one wich is mtreci and torsion free. We define it as being*

$$nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\sigma} V^\sigma \tag{1}$$

where the christofells symbols are

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \tag{2}$$

2.2 spin connexion, vielbein and flatspace

3 Lagrangian field theory

3.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantities called Lagrangian density, which will often refered to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu}) dxdydz \tag{3}$$

Now using esintein convention we can rewritte the action in a new light :

$$S = \int \mathcal{L} dx^\mu \tag{4}$$

3.2 General exemple for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical prescription for scalar and vector fields, but no general rule exists for other cases each field type requires its own treatment. For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.
So for a scalar field we have :

$$\boxed{\mathcal{L} = -\frac{1}{2}\partial_\mu\Phi\partial^\mu\Phi} \tag{5}$$

and for a vectorial field we have

$$\boxed{\mathcal{L} = -\frac{1}{-}F_{\mu\nu}F^{\mu\nu}} \tag{6}$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A^\nu - \partial_\nu A^\mu$$

In a vaccum the einstein equation give us

$$\boxed{\mathcal{L} = R}$$

where R is ricci scalar finally for the lagrangian of a two form fields whe will look in [5] where it is given by the equation :

$$\boxed{\mathcal{L} = g^{\hat{\mu}\mu'}g^{\nu\nu'}g^{\rho\rho'}H_{\mu\nu\rho}H_{\mu'\nu'\rho'}} \tag{7}$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + cyc.perm$$

4 Reduction of hilbert-einstein Action in D+E dimension

In this section we perform a Kaluza-Klein reduction. We consider a universe living in $D+E$ dimensions, with $D = 4$, and we try to determine how an observer confined to the D-dimensional world perceives the extra dimensions.

4.1 Hypothesis

We assume that none of the fields depend on the internal coordinates, i.e. for any field $\partial_\alpha A^{\hat{\mu}} = 0$. It is also worth noting that Kaluza originally performed his reduction on a circle S^1; here, instead, we consider a general internal space of dimension E.

4.2 Rewriting the action

We start from the usual Einstein-Hilbert action, the only difference being that we now work in $D+E$ dimensions. Following [1], this gives

$$\boxed{S = \frac{-1}{\kappa^2}\int dx^D \int \frac{dy^E}{\rho(E)}\hat{V}\hat{R}} \tag{8}$$

where $\rho(E)$ is a normalisation factor introduced in (8), corresponding in practice to the invariant volume of the internal space; κ is fixed such that $\kappa^2/4\pi$ is the usual Newton constant.
Working instead with the spin-connexion formalism, the action can be rewritten in a more transparent form:

$$S = \frac{-1}{\kappa^2}\int dx^D \int \frac{dy^E}{\rho(E)}\hat{V}(\hat{\omega}^{\hat{r}}{}_{\hat{s}\hat{t}}\hat{\omega}^{\hat{s}\hat{t}}{}_{\hat{r}} + \hat{\omega}^{\hat{r}}{}_{\hat{r}\hat{t}}\hat{\omega}^{\hat{s}}{}_{\hat{s}}{}^{\hat{t}}) \tag{9}$$

where the spin connexion is given by

$$\hat{\omega}_{\hat{t}\hat{r}\hat{s}} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}} \tag{10}$$

and the $\hat{\Omega}$ are expressed in terms of the inverse vielbein as

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2}(\hat{V}^{\hat{\mu}}_{\hat{r}}\hat{V}^{\hat{\nu}}_{\hat{s}} - \hat{V}^{\hat{\mu}}_{\hat{s}}\hat{V}^{\hat{\nu}}_{\hat{r}})\partial_{\hat{\nu}}\hat{V}_{\hat{\mu}\hat{t}} \tag{11}$$

4.3 vielbein ansatz

Let us now begin the actual computation. Since all the information about the curved-space metric is contained in the vielbein, we look for an ansatz directly on it. Moreover, following [1], the local Lorentz invariance in $D+E$ dimensions allows us to choose a triangular parametrisation. Our ansatz therefore reads

$$\boxed{\hat{V}^{\hat{r}}_{\hat{\mu}} = \begin{pmatrix} \delta^\gamma V^r_\mu & 2\kappa A^\alpha_\mu \Phi^a_\alpha \\ 0 & \Phi^a_\alpha \end{pmatrix}} \tag{12}$$

In this expression, $\delta = \det\Phi$, and γ is a free parameter corresponding to an arbitrary Weyl rescaling of the D-dimensional vielbein. The vielbein in (25) thus contains the D-dimensional vielbein (i.e. the graviton), E vector fields A^α_μ, and E^2 scalar fields Φ^a_α. However, because of the local Lorentz invariance in the internal directions, only $\frac{1}{2}E(E+1)$ of these scalars actually correspond to propagating degrees of freedom.

4.3.1 Gauge transformations of the fields

Using the fact that the theory is invariant under the classical transformation $x'^{\mu} = x^{\mu} + \xi^{\mu}(x)$ [1], which can be rewritten as

$$\delta \hat{V}^{\hat{r}}_{\hat{\mu}} = \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}^{\hat{r}}_{\hat{\mu}} + \partial_{\hat{\mu}} \xi^{\hat{\rho}} \hat{V}^{\hat{r}}_{\hat{\rho}} \tag{13}$$

we now apply this transformation to each block of the vielbein, in order to see whether it corresponds to some familiar type of field. Let us start with the block $\hat{V}^{r}_{\mu}$:

$$\begin{aligned}
\delta \hat{V}^{r}_{\mu} &= \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}^{r}_{\mu} + \partial_{\mu} \xi^{\hat{\rho}} \hat{V}^{r}_{\hat{\rho}} \\
&= \xi^{\hat{\rho}} \partial_{\hat{\rho}} \hat{V}^{r}_{\mu} + \partial_{\mu} \xi^{\nu} \hat{V}^{r}_{\nu} + \underbrace{\partial_{\mu} \xi^{\alpha} \hat{V}^{r}_{\alpha}}_{\text{this is null}} \\
&= \xi^{\hat{\rho}} \partial_{\hat{\rho}} (\delta^{\gamma} V^{r}_{\mu} + \delta^{\gamma} \partial_{\mu} \xi^{\nu} V^{r}_{\nu} \\
&= \delta^{\gamma} (\xi^{\hat{\rho}} \partial_{\hat{\rho}} V^{r}_{\mu} + \partial_{\mu} \xi^{\nu} V^{r}_{\nu}) + \xi^{\hat{\rho}} \partial_{\hat{\rho}} \delta^{\gamma} V^{r}_{\mu}
\end{aligned}$$

We can also redo this computation using a different method:

$$\begin{aligned}
\delta \hat{V}^{r}_{\mu} &= \delta(\delta^{\gamma} V^{r}_{\mu}) \\
&= \delta(\delta^{\gamma}) V^{r}_{\mu} + \delta^{\gamma} \delta(V^{r}_{\mu})
\end{aligned}$$

and by identification we find

$$\delta V^{r}_{\mu} = \xi^{\hat{\rho}} \partial_{\hat{\rho}} V^{r}_{\mu} + \partial_{\mu} \xi^{\nu} V^{r}_{\nu}$$

which is exactly the transformation law of a vector field under a Lie derivative [4]. Applying the same method to the other blocks, we obtain, for Φ^{a}_{α}:

$$\delta \Phi^{a}_{\alpha} = \xi^{\rho} \partial_{\rho} \Phi^{a}_{\alpha}$$

which is exactly the transformation law of a scalar field. Finally, for A^{α}_{μ} we obtain:

$$A^{\alpha}_{\mu} = \xi^{\rho} \partial_{\rho} A^{\alpha}_{\mu} + \partial_{\mu} \xi^{\rho} A^{\alpha}_{\rho} + \frac{\partial_{\mu} \xi^{\alpha}}{2\kappa}$$

which corresponds to the transformation law of a vector field, up to an extra correction term. One may further note that, restricting to ξ^{α} only, we obtain

$$\delta V^{r}_{\mu} = \delta \Phi^{a}_{\alpha} = 0, \tag{14a}$$

$$\delta A^{\alpha}_{\mu} = \frac{\partial_{\mu} \xi^{\alpha}}{2\kappa}. \tag{14b}$$

so that what remains of the D-dimensional diffeomorphisms, once restricted to ξ^{α} independent of the internal coordinates y, corresponds to a set of E abelian gauge symmetries.

4.3.2 Metrics

Looking back at (25), we notice that Φ^{a}_{α} is in fact nothing but the internal vielbein. We can therefore define the internal metric $h_{\alpha\beta} = \Phi^{a}_{\alpha} \delta_{ab} \Phi^{b}_{\beta}$, and finally write down the full $(D+E)$-dimensional metric $\hat{g}_{\hat{\mu}\hat{\nu}}$ as

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} \delta^{2\gamma} g_{\mu\nu} - 4\kappa^{2} A^{\gamma}_{\mu} A_{\nu\lambda} & -2\kappa A_{\mu\beta} \\ -2\kappa A_{\nu\alpha} & -h_{\alpha\beta} \end{pmatrix} \tag{15}$$

and, just as forthrightly, the inverse metric

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} \delta^{2\gamma} g^{\mu\nu} & -2\kappa A^{\mu\beta} \\ -2\kappa A^{\nu\alpha} & -h^{\alpha\beta} + 4\kappa^{2} A^{\beta}_{\rho} A^{\rho\alpha} \end{pmatrix} \tag{16}$$

4.4 Latin connexion computation

We now need to compute some of the spin-connexion components. To see which ones are actually required, let us rewrite the action in terms of blocks of non-hatted (flat) indices:

$$\begin{aligned}
S = -\frac{1}{4\kappa^{2}} \int d^{D}x \int \frac{d^{E}y}{\rho(E)} \hat{V} \Big[& \hat{\omega}^{a}{}_{bc} \hat{\omega}^{bc}{}_{a} + \hat{\omega}^{a}{}_{bt} \hat{\omega}^{bt}{}_{a} + \hat{\omega}^{a}{}_{sc} \hat{\omega}^{sc}{}_{a} + \hat{\omega}^{a}{}_{st} \hat{\omega}^{st}{}_{a} + \hat{\omega}^{r}{}_{bc} \hat{\omega}^{bc}{}_{r} + \hat{\omega}^{r}{}_{bt} \hat{\omega}^{bt}{}_{r} + \hat{\omega}^{r}{}_{sc} \hat{\omega}^{sc}{}_{r} \\
& \hat{\omega}^{r}{}_{st} \hat{\omega}^{st}{}_{r} + \hat{\omega}^{a}{}_{ac} \hat{\omega}^{b}{}_{b}{}^{c} + \hat{\omega}^{a}{}_{at} \hat{\omega}^{b}{}_{b}{}^{t} + \hat{\omega}^{a}{}_{ac} \hat{\omega}^{s}{}_{s}{}^{c} + \hat{\omega}^{a}{}_{at} \hat{\omega}^{s}{}_{s}{}^{t} + \hat{\omega}^{r}{}_{rc} \hat{\omega}^{b}{}_{b}{}^{c} + \hat{\omega}^{r}{}_{rt} \hat{\omega}^{b}{}_{b}{}^{t} + \hat{\omega}^{r}{}_{rc} \hat{\omega}^{s}{}_{s}{}^{c} + \hat{\omega}^{r}{}_{rt} \hat{\omega}^{s}{}_{s}{}^{t} \Big]
\end{aligned} \tag{17}$$

In order to minimise the number of connexion components we need to compute, we use the identity

$$\omega_{rst} = -\omega_{rts} \tag{18}$$

Furthermore, for any pair of up/down indices, one can perform the following manipulation:

$$\omega_{rs}{}^{t} = \eta^{tn}\omega_{rsn} = -\eta^{tn}\omega_{rns} = -\omega_{r}{}^{t}{}_{s}$$

This gives a convenient trick to reduce, as much as possible, the number of connexion components that must be computed, allowing us to rewrite the action above as

$$\boxed{\begin{aligned} S = -\frac{1}{4\kappa^2}\int d^Dx \int \frac{d^Ey}{\rho(E)}\hat{V}\Big[& \hat{\omega}^a{}_{bc}\hat{\omega}^{bc}{}_a + \hat{\omega}^a{}_{bt}\hat{\omega}^{bt}{}_a + \hat{\omega}^a{}_{sc}\hat{\omega}^{sc}{}_a + \hat{\omega}^a{}_{st}\hat{\omega}^{st}{}_a - \hat{\omega}^r{}_{bc}\hat{\omega}^b{}_r{}^c - \hat{\omega}^r{}_{tb}\hat{\omega}^{bt}{}_r - \hat{\omega}^r{}_{sc}\hat{\omega}^{cs}{}_r \\ & + \hat{\omega}^r{}_{st}\hat{\omega}^{st}{}_r + \hat{\omega}^a{}_{ac}\hat{\omega}^b{}_b{}^c + \hat{\omega}^a{}_{ta}\hat{\omega}^{bt}{}_b + \hat{\omega}^a{}_{ac}\hat{\omega}^s{}_s{}^c - \hat{\omega}^a{}_{ta}\hat{\omega}^s{}_s{}^t + \hat{\omega}^r{}_{rc}\hat{\omega}^b{}_b{}^c - \hat{\omega}^r{}_{rt}\hat{\omega}^{bt}_b + \hat{\omega}^r{}_{rc}\hat{\omega}^s{}_s{}^c + \hat{\omega}^r{}_{rt}\hat{\omega}^s{}_s{}^t \Big] \end{aligned}} \tag{19}$$

It then becomes clear that we only need to compute six components of ω, namely

$$\hat{\omega}_{pmn},\ \hat{\omega}_{pma},\ \hat{\omega}_{pab},$$
$$\hat{\omega}_{cmn},\ \hat{\omega}_{cma},\ \hat{\omega}_{cab},$$

One can also check that computing these six components only requires the following eight components of Ω:

$$\begin{aligned} &\hat{\Omega}_{pmn}, \\ &\hat{\Omega}_{pma},\ \hat{\Omega}_{pam},\ \hat{\Omega}_{apm}, \\ &\hat{\Omega}_{mab},\ \hat{\Omega}_{amb},\ \hat{\Omega}_{abm}, \\ &\hat{\Omega}_{abc}, \end{aligned}$$

Using (10), (11) and (25), we can now set out a clear strategy: first compute $\hat{V}_{\hat{r}}^{\hat{\mu}}$ and $\hat{V}_{\hat{\mu}\hat{r}}$, then the Ω's, and finally the ω's. The inverse vielbein is easy to obtain, since

$$\hat{V}_{\hat{\mu}}^{\hat{s}}\hat{V}_{\hat{r}}^{\hat{\mu}} = \delta_{\hat{s}}^{\hat{r}}$$

so that

$$\hat{V}_{\hat{\mu}}^{\hat{r}} = \begin{pmatrix} \delta^{-\gamma}V_r^{\mu} & -2\kappa\delta^{-\gamma}A_r^{\alpha} \\ 0 & \Phi_a^{\alpha} \end{pmatrix} \tag{20}$$

Similarly, for $\hat{V}_{\hat{\mu}\hat{r}}$, using

$$\hat{V}_{\hat{r}}^{\hat{\mu}}\hat{V}_{\hat{\mu}\hat{s}} = \eta_{\hat{r}\hat{s}}$$

we obtain

$$\hat{V}_{\hat{\mu}\hat{s}} = \begin{pmatrix} \delta^{\gamma}V_{\mu s} & 2\kappa A_{\mu}^{\alpha}\Phi_{\alpha a} \\ 0 & \Phi_{\alpha a} \end{pmatrix} \tag{21}$$

Let us now turn to the computation of the Ω's themselves. Starting with $\hat{\Omega}_{pmn}$:

$$\begin{aligned} \hat{\Omega}_{pmn} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_m^{\nu} - \hat{V}_m^{\hat{\mu}}\hat{V}_p^{\nu})\partial_\nu\hat{V}_{\hat{\mu}n} \\ &= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_m^{\nu} - \hat{V}_m^{\mu}\hat{V}_p^{\nu})\partial_\nu\hat{V}_{\mu n} \\ &= \frac{1}{2}(\hat{V}_p^{\mu}\hat{V}_m^{\nu} - \hat{V}_m^{\mu}\hat{V}_p^{\nu})\partial_\nu\hat{V}_{\mu n} \\ &= \frac{\delta^{-2\gamma}}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_\nu(\delta^{\gamma}V_{\mu n}) \\ &= \frac{\delta^{-2\gamma}}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_\nu(\delta^{\gamma})V_{\mu n} + \delta^{\gamma}\partial_\nu(V_{\mu n})) \\ &= \frac{\delta^{-2\gamma}}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\gamma\partial_\nu(\delta)\delta^{\gamma-1}V_{\mu n} + \delta^{\gamma}\partial_\nu(V_{\mu n})) \\ &= \frac{\delta^{-\gamma}}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\gamma\partial_\nu(\delta)\delta^{-1}V_{\mu n} + \partial_\nu(V_{\mu n})) \\ &= \frac{\delta^{-\gamma}}{2}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\gamma\partial_\nu(\log\delta)V_{\mu n} + \partial_\nu(V_{\mu n})) \end{aligned}$$

For $\hat{\Omega}_{pma}$, we find:

$$\begin{aligned}
\hat{\Omega}_{pma} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_m^{\nu} - \hat{V}_m^{\hat{\mu}}\hat{V}_p^{\nu})\partial_\nu \hat{V}_{\hat{\mu}a} \\
&= -\kappa\delta^{-2\gamma}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_\nu(A_\mu^\alpha \Phi_{a\alpha}) + \kappa\delta^{-\gamma}(A_p^\alpha V_m^\nu - \hat{A}_m^\alpha V_p^\nu)\partial_\nu(\Phi_{a\alpha}) \\
&= -\kappa\delta^{-2\gamma}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_\nu(A_\mu^\alpha)\Phi_{a\alpha} + \partial_\nu(\Phi_{a\alpha})A_\mu^\alpha) + \kappa\delta^{-\gamma}(A_p^\alpha V_m^\nu - \hat{A}_m^\alpha V_p^\nu)\partial_\nu(\Phi_{a\alpha}) \\
&= -\kappa\delta^{-2\gamma}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})(\partial_\nu(A_\mu^\alpha)\Phi_{a\alpha} + \partial_\nu(\Phi_{a\alpha})A_\mu^\alpha) + (A_p^\alpha V_m^\nu - \hat{A}_m^\alpha V_p^\nu)\partial_\nu(\Phi_{a\alpha}) \\
&= -\kappa\delta^{-2\gamma}(V_p^{\mu}V_m^{\nu} - V_m^{\mu}V_p^{\nu})\partial_\nu(A_\mu^\alpha)\Phi_{a\alpha} \\
&= \kappa\delta^{-2\gamma}(F^\alpha{}_{pm}\Phi_{a\alpha} - \partial_\nu(V_p^\mu V_m^\nu)A_\mu^\alpha\Phi_{a\alpha} + \partial_\nu(V_p^\nu V_m^\mu)A_\mu^\alpha\Phi_{a\alpha}) \\
&= \kappa\delta^{-2\gamma}F^\alpha{}_{pm}\Phi_{a\alpha}
\end{aligned}$$

For $\hat{\Omega}_{pam}$:

$$\begin{aligned}
\hat{\Omega}_{pam} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_\nu \hat{V}_{\hat{\mu}m} \\
&= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_\nu \hat{V}_{\hat{\mu}m} \\
&= 0
\end{aligned}$$

The same result holds for $\hat{\Omega}_{apm}$.
For $\hat{\Omega}_{pab}$:

$$\begin{aligned}
\hat{\Omega}_{pab} &= \frac{1}{2}(\hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu} - \hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu})\partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{-\delta^{-\gamma}}{2}\hat{V}_a^{\hat{\mu}}V_p^{\nu}\partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha V_p^\nu \partial_\nu \Phi_{\alpha b} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha V_p^\nu \partial_\nu (h_{\alpha\beta}\Phi_b^\beta) \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} + \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha V_p^\nu h_{\alpha\beta}\partial_\nu \Phi_b^\beta \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} + \frac{\delta^{-\gamma}}{2}\Phi_{\beta a} V_p^\nu \partial_\nu \Phi_b^\beta \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} + \frac{\delta^{-\gamma}}{2}V_p^\nu \partial_\nu (\underbrace{\Phi_{\beta a}\Phi_b^\beta}_{=\delta_{ab}\text{i.e is constant}constant}) - \Phi_b^\beta V_p^\nu \partial_\nu \Phi_{\beta a} \\
&= \frac{\delta^{-\gamma}}{2}\Phi_a^\alpha \Phi_b^\beta V_p^\nu \partial_\nu h_{\alpha\beta} - \Phi_b^\beta V_p^\nu \partial_\nu \Phi_{\beta a}
\end{aligned}$$

For $\hat{\Omega}_{apb}$:

$$\begin{aligned}
\hat{\Omega}_{apb} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu} - \hat{V}_p^{\hat{\mu}}\hat{V}_a^{\nu})\partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2}\hat{V}_a^{\hat{\mu}}\hat{V}_p^{\nu}\partial_\nu \hat{V}_{\hat{\mu}b} \\
&= \frac{1}{2}\hat{V}_a^{\alpha}\hat{V}_p^{\nu}\partial_\nu \hat{V}_{\alpha b} \\
&= \frac{-\delta^{-\gamma}}{2}\Phi_a^\alpha V_p^\nu \partial_\nu \Phi_{\alpha b}
\end{aligned}$$

For $\hat{\Omega}_{abp}$:

$$\begin{aligned}
\hat{\Omega}_{abp} &= \frac{1}{2}(\hat{V}_a^{\hat{\mu}}\hat{V}_b^{\nu} - \hat{V}_b^{\hat{\mu}}\hat{V}_a^{\nu})\partial_\nu \hat{V}_{\hat{\mu}p} \\
&= 0
\end{aligned}$$

The same holds for Ω_{abc}. Using (10), we can then obtain the ω's straightforwardly:

$$\hat{\omega}_{pmn} = \delta^{-\gamma}[\omega_{pmn} + \gamma(\eta_{pm}V_n^{\lambda} - \eta_{pn}V_m^{\lambda})\partial_\lambda], \tag{22a}$$
$$\hat{\omega}_{pma} = \kappa\delta^{-2\gamma}F^{\alpha}_{pm}\Phi_{\alpha a}, \tag{22b}$$
$$\hat{\omega}_{pab} = \frac{1}{2}\delta^{-\gamma}\Phi^{\alpha}_{a}V^{\mu}_{p}\partial_\mu\Phi_{\alpha\beta} - (a \leftrightarrow b), \tag{22c}$$
$$\hat{\omega}_{cmn} = -\kappa\delta^{-2\gamma}F^{\alpha}_{mn}\Phi_{\alpha c}, \tag{22d}$$
$$\hat{\omega}_{cab} = 0 \tag{22e}$$

where $F^{\alpha}_{\mu\nu} = \partial_\mu A^{\alpha}_{\nu} - \partial_\nu A^{\alpha}_{\mu}$.
The gauge invariance of these expressions is then manifest.

4.5 finalisation

Now that all this work has been done we can finally begin the computation of the action which gave us

$$S = \frac{-1}{4\kappa^2}\int dx^D \int \frac{dy^E}{\rho(E)} V\delta^{\gamma D+1}(\frac{1}{4}\delta^{-2\gamma}g^{\rho\lambda}\partial_\lambda h_{\alpha\beta}\partial_\rho h^{\alpha\beta} + \kappa^2\delta^{-2\gamma}F^{\mu\nu\alpha}F^{\beta}_{\mu\nu}h_{\alpha\beta}+$$
$$\delta^{-2\gamma}(\omega^{r}_{st}\omega^{st}_{r} + \gamma^2(2-D)g^{\lambda\rho}\partial_\rho\log\delta\partial_\lambda\log\delta) + \delta^{-2\gamma}(\omega^{r}_{rt}\omega^{s\ \ t}_{\ s}))$$

or rewriten in a nicer way

$$S = \int dx^D \int \frac{dy^E}{\rho(E)} V\delta^{\gamma D+1}(\frac{-1}{16}\delta^{-2\gamma}g^{\rho\lambda}\partial_\lambda h_{\alpha\beta}\partial_\rho h^{\alpha\beta} + \frac{-1}{4}\delta^{-2\gamma}F^{\mu\nu\alpha}F^{\beta}_{\mu\nu}h_{\alpha\beta}+$$
$$\frac{-1}{4\kappa^2}\delta^{-2\gamma}\gamma^2(2-D)g^{\lambda\rho}\partial_\rho\log\delta\partial_\lambda\log\delta + \frac{-1}{4\kappa^2}\delta^{-2\gamma}(\omega^{r}_{st}\omega^{st}_{r} + \omega^{r}_{rt}\omega^{s\ \ t}_{\ s}))$$

so now we can identify R the D-dimensoin einstein action and see the factor $\delta^{\gamma(D-2)+1}$ the good thing is that γ is a free variable so I can just say that

$$\boxed{\begin{aligned} S = \int dx^D V(-\frac{1}{4\kappa^2}R + \frac{-1}{4}\delta^{\frac{2}{D-2}}F^{\mu\nu\alpha}F^{\beta}_{\mu\nu}h_{\alpha\beta}+ \\ -\frac{1}{16}g^{\rho\lambda}\partial_\lambda h_{\alpha\beta}\partial_\rho h^{\alpha\beta} + \frac{1}{4\kappa^2(D-2)}g^{\lambda\rho}\partial_\rho\log\delta\partial_\lambda\log\delta) \end{aligned}} \tag{23}$$

5 reduction of the new fields

Now that we have seen the classical reduction of the esintein action an intersting exercise we can do is to add two new field namely the dilaton field and a second rank antysemtric tensor as done in [5] We will once again follow the same path for the reduction nowing that for this part [5] did follow the instruction og [1]. The main change in both paper is that [5] choose $\gamma = 0$ and not $\gamma = -\frac{-1}{D-2}$, which as said earlier is just a choice who do not carrie a lot of importance. [5] also chose to ignore the κ which gave us a modified action in comparions to our last result. Finally contrary to [1] ,[5] didn"t choose the (+,-,-,-,-,...) signature for the tangeant space metric but (-,+,+,+,+,...), so there will be some change of singe in the action. Once everything is took into account we get the new reduced action

$$\boxed{S = \int dx^D\sqrt{-g}\delta(R - \frac{1}{4}F^{\mu\nu\alpha}F^{\beta}_{\mu\nu}h_{\alpha\beta} + -g^{\rho\lambda}\partial_\lambda h_{\alpha\beta}\partial_\rho h^{\alpha\beta})} \tag{24}$$

and of course we can't forget to rewrite the vielbein and the high dimension metric

$$\hat{V}^{\hat{r}}_{\hat{\mu}} = \begin{pmatrix} V^{r}_{\mu} & A^{\alpha}_{\mu}\Phi^{a}_{\alpha} \\ 0 & \Phi^{a}_{\alpha} \end{pmatrix} \quad \text{and} \quad \hat{V}^{\hat{\mu}}_{\hat{r}} = \begin{pmatrix} V^{\mu}_{r} & -A^{\alpha}_{r} \\ 0 & \Phi^{\alpha}_{a} \end{pmatrix} \tag{25}$$

and the metric

$$\hat{g}_{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g_{\mu\nu} + A^{\gamma}_{\mu}A_{\nu\lambda} & A_{\mu\beta} \\ A_{\nu\alpha} & h_{\alpha\beta} \end{pmatrix} \quad \text{and} \quad \hat{g}^{\hat{\mu}\hat{\nu}} = \begin{pmatrix} g^{\mu\nu} & -A^{\mu\beta} \\ -A^{\nu\alpha} & -h^{\alpha\beta} + A^{\beta}_{\rho}A^{\rho\alpha} \end{pmatrix} \tag{26}$$

As a final step before begining we can see that

$$\sqrt{-g} = det\hat{V}^{\hat{r}}_{\hat{\mu}} = detV^{r}_{\mu}det\Phi^{a}_{\alpha} = \sqrt{-g}\sqrt{h} \tag{27}$$

5.1 scalar field reduction

Now let's add the dilaton field by modifying like this the original action

$$S = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{-\hat{\phi}} \delta(\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial_{\hat{\nu}} \hat{\phi}) \tag{28}$$

We can then do what we did with the [1] and get and other manipulation

$$\begin{aligned} S = \int dx^D \sqrt{-g}\sqrt{h} e^{-\hat{\phi}} \delta(R - \frac{1}{4} F^{\mu\nu\alpha} F^{\beta}_{\mu\nu} h_{\alpha\beta} + \\ - g^{\rho\lambda} \partial_\lambda h_{\alpha\beta} \partial_\rho h^{\alpha\beta} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial_{\hat{\nu}} \hat{\phi})) \end{aligned} \tag{29}$$

5.2 Two form fiel reduction

5.2.1 The reduction

For the two form we will reduce it apart from the two other field, this those not cause any problem as the action is in effect linear.
We can rewrite (7) as

$$\begin{aligned} \mathcal{L}_{\hat{\mathcal{B}}} &= \frac{-1}{12} (g^{\hat{\mu}\hat{\mu}'} g^{\hat{\nu}\hat{\nu}'} g^{\hat{\rho}\hat{\rho}'} \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \hat{H}_{\hat{\mu}'\hat{\nu}'\hat{\rho}'}) \\ &= \frac{-1}{12} (\hat{\eta}^{\hat{r}\hat{r}'} \hat{\eta}^{\hat{s}\hat{s}'} \hat{\eta}^{\hat{t}\hat{t}'} \hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}_{\hat{r}'\hat{s}'\hat{t}'}) \\ &= \frac{-1}{12} (\hat{H}_{\hat{r}\hat{s}\hat{t}} \hat{H}^{\hat{r}\hat{s}\hat{t}}) \\ &= \frac{-1}{12} (\hat{H}_{rst}\hat{H}^{rst} + \hat{H}_{rsc}\hat{H}^{rsc} + \hat{H}_{rbt}\hat{H}^{rbt} + \hat{H}_{rbc}\hat{H}^{rbc} + \hat{H}_{ast}\hat{H}^{ast} + \hat{H}_{asc}\hat{H}^{asc} + \hat{H}_{abs}\hat{H}^{abs} + \hat{H}_{abc}\hat{H}^{abc}) \end{aligned}$$

however thanks to to the fact that $\hat{B}_{\hat{\mu}\hat{\nu}}$ is antysymetric one can easly show that this imply that $\hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}}$ is antysemtric for any neighbor index. so we obtaint

$$\mathcal{L}_{\hat{\mathcal{B}}} = \frac{-1}{12} (\hat{H}_{rst}\hat{H}^{rst} + 3\hat{H}_{rsc}\hat{H}^{rsc} + 3\hat{H}_{asc}\hat{H}^{asc} + \hat{H}_{abc}\hat{H}^{abc}) \tag{30}$$

and finnaly the action of B is writen as

$$\boxed{S_{\hat{b}} = -\int dx^D \sqrt{-g} e^{-\phi} \left\{ \frac{1}{12} H_{\alpha\beta\gamma} H^{\alpha\beta\gamma st} + \frac{1}{4} H_{\mu\alpha\beta} H^{\mu\alpha\beta} + \frac{1}{4} H_{\mu\nu\alpha} H^{\mu\nu\alpha} + \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \right\}} \tag{31}$$

You might have remarked that there is no more hat it is due to the fact that we have done a simple manipulation that I will show rigt there

$$T^\mu = V^\mu_r \hat{V}^r_{\hat{\mu}} \hat{T}^{\hat{\mu}} = \hat{V}^\alpha \tag{32a}$$
$$T^\alpha = V^\alpha_a \hat{V}^a_{\hat{\mu}} \hat{T}^{\hat{\mu}} = \hat{V}^\alpha + A^{(1)\alpha}_\mu \hat{V}^\mu \tag{32b}$$
$$T_\mu = V^r_\mu \hat{V}^{\hat{\mu}}_r \hat{T}_{\hat{\mu}} = \hat{V}_\mu - A^{(1)\alpha}_\mu \hat{V}_\alpha \tag{32c}$$
$$T_\alpha = V^a_\alpha \hat{V}^{\hat{\mu}}_a \hat{T}_{\hat{\mu}} = \hat{V}_\alpha \tag{32d}$$
$$\tag{32e}$$

so with that we obtaint eq:31
Now we will compute each componant of non hated H in function of hated H
to begin slowly as we have seen above $\hat{B}_{\alpha\beta} = B_{\alpha\beta}$ and if we use the y independence we get $H_{\alpha\beta\gamma} = 0$
We continue with $H_{\mu\alpha\beta}$

$$\begin{aligned} H_{\mu\alpha\beta} &= V^r_\mu \hat{V}^{\hat{\mu}}_r \hat{H}_{\hat{\mu}\alpha\beta} \\ &= \partial_\mu B_{\alpha\beta} \end{aligned} \tag{33}$$

those two were quite easy howver the next two are going to be more troublesome, but we are going to introduce new object
Now for $H_{\mu\nu\beta}$

$$\begin{aligned} H_{\mu\alpha\beta} &= V^r_\mu V^s_\nu \hat{V}^{\hat{\mu}}_r \hat{V}^{\hat{\nu}}_s \hat{H}_{\hat{\mu}\hat{\nu}\alpha} \\ &= \hat{H}_{\mu\nu\alpha} - (A^{(1)\alpha}_\mu \hat{H}_{\alpha\nu\rho} + A^{(1)\alpha}_\rho \hat{H}_{\mu\alpha\nu}) \\ &= \partial_\mu \hat{B}_{\nu\alpha} + \partial_\nu \hat{B}_{\alpha\mu} - A^{(1)\beta}_\mu \partial_\nu \hat{B}_{\alpha\beta} - A^{(1)\beta}_\nu \partial_\mu \hat{B}_{\beta\alpha} \\ &= \partial_\mu \hat{B}_{\nu\alpha} + \partial_\nu \hat{B}_{\alpha\mu} - \partial_\nu (A^{(1)\beta}_\mu \hat{B}_{\alpha\beta}) + \partial_\nu A^{(1)\beta}_\mu \hat{B}_{\alpha\beta} - \partial_\mu (A^{(1)\beta}_\nu \hat{B}_{\beta\alpha}) + \partial_\mu A^{(1)\beta}_\nu \hat{B}_{\beta\alpha} \\ &= \partial_\mu \hat{B}_{\nu\alpha} - \partial_\nu \hat{B}_{\mu\alpha} - \partial_\nu (A^{(1)\beta}_\mu \hat{B}_{\alpha\beta}) + \partial_\nu A^{(1)\beta}_\mu \hat{B}_{\alpha\beta} + \partial_\mu (A^{(1)\beta}_\nu \hat{B}_{\alpha\beta}) - \partial_\mu A^{(1)\beta}_\nu \hat{B}_{\alpha\beta} \\ &= \partial_\mu (\hat{B}_{\nu\alpha} + A^{(1)\beta}_\nu \hat{B}_{\alpha\beta}) - \partial_\nu (\hat{B}_{\mu\alpha} + A^{(1)\beta}_\mu \hat{B}_{\alpha\beta}) - \hat{B}_{\alpha\beta} (\partial_\mu A^{(1)\beta}_\nu - \partial_\mu A^{(1)\beta}_\nu) \\ &= F^{(2)}_{\mu\nu\alpha} - \hat{B}_{\alpha\beta} F^{(1)\beta}_{\mu\nu} \end{aligned} \tag{34}$$

and finnaly for $H_{\mu\nu\rho}$ we are just going to use the exam same technic as before which gave us

$$\begin{aligned} H_{\mu\nu\rho} &= V^r_\mu V^s_\nu V^t_\rho \hat{V}^{\hat{\mu}}_r \hat{V}^{\hat{\nu}}_s \hat{V}^{\hat{\rho}}_t \hat{H}_{\hat{\mu}\hat{\nu}\hat{\rho}} \\ &= \hat{H}_{\mu\nu\rho} - A^{(1)\alpha}_\mu \hat{H}_{\alpha\nu\rho} - A^{(1)\alpha}_\nu \hat{H}_{\alpha\rho\mu} - A^{(1)\alpha}_\rho \hat{H}_{\alpha\mu\nu} + A^{(1)\alpha}_\mu A^{(1)\beta}_\nu \hat{H}_{\alpha\beta\rho} + A^{(1)\alpha}_\nu A^{(1)\beta}_\rho \hat{H}_{\alpha\beta\mu} + A^{(1)\alpha}_\rho A^{(1)\beta}_\mu \hat{H}_{\alpha\beta\nu} \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2}(A^{(1)\alpha}_\mu F^{(2)}_{\nu\rho\alpha} + A^{(2)}_{\mu\alpha} F^{(1)}_{\nu\rho}) + \partial_\nu B_{\rho\mu} - \frac{1}{2}(A^{(1)\alpha}_\nu F^{(2)}_{\rho\mu\alpha} + A^{(2)}_{\nu\alpha} F^{(1)}_{\rho\nu}) + \partial_\rho B_{\mu\nu} - \frac{1}{2}(A^{(1)\alpha}_\rho F^{(2)}_{\mu\nu\alpha} + A^{(2)}_{\rho\alpha} F^{(1)}_{\mu\nu}) \end{aligned} \tag{35}$$

where

$$B_{\mu\nu} = \hat{B}_{\mu\nu} + \frac{1}{2}A^{(1)\alpha}_\mu A^{(2)}_{\nu\alpha} - \frac{1}{2}A^{(1)\alpha}_\nu A^{(2)}_{\mu\alpha} - A^{(1)\alpha}_\mu B_{\alpha\beta} A^{(2)\beta}_\nu \tag{36}$$

5.2.2 Complete reduced action

To recap we have the complete reduced action in the form

$$S = \int dx^D \sqrt{(-g)} e^{-\phi} \mathcal{L} \tag{37}$$

Where we decompose $\mathcal{L}$ as $\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3 + \mathcal{L}_4$, and we pose

$$\boxed{\begin{aligned} &\mathcal{L}_1 = R + g^{\mu\nu}\partial_\mu\Phi\partial_\nu\Phi \\ &\mathcal{L}_2 = \frac{1}{4}g^{\mu\nu}(\partial_\mu h_{\alpha\beta}\partial_\nu G^{\alpha\beta} - h^{\alpha\beta}h^{\gamma\delta}\partial_\mu B_{\alpha\gamma}\partial_\nu B_{\beta\delta}) \\ &\mathcal{L}_3 = -\frac{1}{4}g^{\mu\rho}g^{\nu\lambda}(h_{\alpha\beta}F^{(1)\alpha}_{\mu\nu}F^{(1)\beta}_{\rho\lambda} + h^{\alpha\beta}H_{\mu\nu\alpha}H_{\rho\lambda\beta}) \\ &\mathcal{L}_4 = -\frac{1}{12}H_{\mu\nu\rho}H^{\mu\nu\rho} \end{aligned}} \tag{38}$$

5.2.3 Observation of gauge

Now as in part 4.3.1 we are going to observe the gauge rule to be sure that every object transorm as it should so let's begin with A^2 which should transform with the lie derivative and under the new transformations of $B_{\mu\nu} \to B_{\mu\nu} + \partial_m u\Lambda_\nu$, so let's begin with $A^{(2)}_{\mu\alpha}$

$$\begin{aligned} \delta A^{(2)}_{\mu\alpha} &= \delta(\hat{B}_{\mu\alpha} + B_{\alpha\beta}A^{(1)\beta}_\mu) \\ &= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta}A^{(1)\beta}_\mu) \\ &= \delta(\hat{B}_{\mu\alpha}) + \delta(B_{\alpha\beta})A^{(1)\beta}_\mu + B_{\alpha\beta}\delta(A^{(1)\beta}_\mu) \\ &= L_\xi\hat{B}_{\mu\alpha} + \partial_\mu\Lambda^{(2)}_\alpha + A^{(\beta)}_\mu L_\xi\hat{B}_{\alpha\beta} + \hat{B}_{\alpha\beta}\partial_\mu\Lambda^{(1)}_\alpha \\ &= L_\xi\hat{B}_{\mu\alpha} + \partial_\mu\Lambda^{(2)}_\alpha + L_\xi(A^{(\beta)}_\mu\hat{B}_{\alpha\beta}) \\ &= \partial_\mu\Lambda^{(2)}_\alpha + L_\xi A^{(2)}_{\mu\alpha} \end{aligned} \tag{39}$$

so we have the right transformation for this object
Doing the same thing with $B_{\mu\nu}$ we come under the transformation

$$\delta B_{\mu\nu} = \frac{1}{2}(\Lambda^{(1)\alpha}F^{(2)}_{\mu\nu\alpha} + \lambda^{(2)}_\alpha F^{(1)\alpha}_{\mu\nu}) \tag{40}$$

wich is exactly the good terme to compensate the addional term for the transformation of $H_{\mu\nu\rho}$

6 Observation of new symmetry

Now for this part we are going to explore a very important part of physics research nowaday which is symetrie. In our context it will allow us to make equivalence between multpiple theory as whole. But firstly we have to see the obvious symetrie that we have seen in the non-reduced version of the Action which is the SL(E,R). But what will interss us in the reduction is that it does not create nor destroy any symetrie and if we observe a new symetrie in the reduced form that mean that we have effectively discoverd a new symtrie in the unreduced forme. and following the [5] we observe an O(E,E) symetrie. let's define it and then observe if it is true so the definition.

Definition 5 *Any matrices M is in O(E,E) if it is a square $E \times E$ matrices and if for the matrices*

$$\zeta = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

We can use this matrices as a metric for the groupe O(E,E) as it has d -1 eigenvalues and d 1 eigenvalues, in a basis rotated from the one with diagonal metric.
the matrices M verify

$$\boxed{M^T \zeta M = \zeta}$$

We will now define a new matrices M which is

$$M = \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \tag{41}$$

Of course we will verify that this true

$$\begin{aligned} M^T \zeta M &= \begin{pmatrix} h^{-1} & Bh^{-1} \\ -h^{-1}B & h - Bh^{-1}B \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} Bh^{-1} & h^{-1} \\ h - Bh^{-1}B & -h^{-1}B \end{pmatrix} \begin{pmatrix} h^{-1} & -h^{-1}B \\ Bh^{-1} & h - Bh^{-1}B \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \zeta \end{aligned}$$

so we confirm that $M \in O(E,E)$ Now that we have that in mind let's retook (38) and verify that each $\mathcal{L}$ is infact O(E,E) invariant
For $\mathcal{L}_1$ the invzriance is obvious as $g_{\mu\nu}$ and Φ are both invariant. Nonwithsandaning we ought to remamber that for Φ each terms in it are note on there self invariant only.
Now the [5] propose a new tactics to observe the invariance of $\mathcal{L}_2$ which is to rewrite everything in matricial terms. we the obtain

$$\mathcal{L}_2 = \frac{1}{4} tr(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B)$$

And now we are going to do a few manipulation

$$\begin{aligned} \mathcal{L}_2 &= \frac{1}{4} tr(\partial_\mu h^{-1} \partial^\mu h + h^{-1} \partial_\mu B h^{-1} \partial^\mu B) \\ &= \frac{1}{8} tr\Big(2\partial_\mu h \partial^\mu h^{-1} + 2h^{-1} \partial_\mu B h^{-1} \partial^\mu B\Big) \\ &= \frac{1}{8} tr\Big(2\partial_\mu h \partial^\mu h^{-1} - 2\partial_\mu (Bh^{-1}B) \partial^\mu h^{-1} + \partial_\mu (Bh^{-1}) \partial^\mu (Bh^{-1}) + \partial_\mu (h^{-1}B) \partial^\mu (h^{-1}B)\Big) \\ &= \frac{1}{8} tr\Big(\partial_\mu (h - Bh^{-1}B) \partial^\mu h^{-1} + \partial_\mu (Bh^{-1}) \partial^\mu (Bh^{-1}) + \partial_\mu (h^{-1}B) \partial^\mu (h^{-1}B) + \partial^\mu h^{-1} \partial_\mu (h - Bh^{-1}B)\Big) \\ &= \frac{1}{8} tr(\partial_\mu M^{-1} \partial^\mu M) \end{aligned} \tag{42}$$

Now we are going to rewrite $\mathcal{L}_3$ by developing the H partial to get

$$\begin{aligned} \mathcal{L}_3 &= -\frac{1}{4} [F^{(1)\alpha}_{\mu\nu} h_{\alpha\beta} F^{(1)\mu\nu\beta} + (F^{(2)}_{\mu\nu\alpha} - B_{\alpha\gamma} F^{(1)\gamma}_{\mu\nu}) h^{\alpha\beta} (F^{(2)\mu\nu}_{\beta} - B_{\beta\delta} F^{(1)\mu\nu\delta})] \\ &= -\frac{1}{4} \mathcal{F}^i_{\mu\nu} (M^{-1})_{ij} \mathcal{F}^{\mu\nu j} \end{aligned} \tag{43}$$

Where $\mathcal{F}^i_{\mu\nu}$ is the 2E-component vector of field strengths

$$\mathcal{F}^i_{\mu\nu} = \begin{pmatrix} F^{(1)\alpha}_{\mu\nu} \\ F^{(2)}_{\mu\nu\alpha} \end{pmatrix} = \partial_\mu \mathcal{A}^i_\nu - \partial_\nu \mathcal{A}^i_\mu \tag{44}$$

Now the O(E,E) invariance is visible if we consider that the vector fields transform linealy to the vector representation of O(E,E), i.e., $\mathcal{A}^i_\mu \to \Omega^i_j \mathcal{A}^j_\mu$ And finnaly we go to $\mathcal{L}_4$ firstly let's rewrite $H_{\mu\nu\rho}$

$$\begin{aligned} H_{\mu\nu\rho} &= \partial_\mu B_{\nu\rho} - \frac{1}{2}(A^{(1)\alpha}_\mu F^{(2)}_{\nu\rho\alpha} + A^{(2)}_{\mu\alpha} F^{(1)}_{\nu\rho}) + \partial_\nu B_{\rho\mu} - \frac{1}{2}(A^{(1)\alpha}_\nu F^{(2)}_{\rho\mu\alpha} + A^{(2)}_{\nu\alpha} F^{(1)}_{\rho\nu}) + \partial_\rho B_{\mu\nu} - \frac{1}{2}(A^{(1)\alpha}_\rho F^{(2)}_{\mu\nu\alpha} + A^{(2)}_{\rho\alpha} F^{(1)}_{\mu\nu}) \\ &= \partial_\mu B_{\nu\rho} - \frac{1}{2} \mathcal{A}^i_\mu \zeta_{ij} \mathcal{F}^j_{\nu\rho} + (cyc.perms) \end{aligned} \tag{45}$$

Now as $\Omega^T \zeta \Omega = \zeta$ the seconds thermes is O(E,E) invariant. And the for the first terme is invariant if we require that $B_{\nu\rho}$ does not transfrom. This demonstrates that the action hides a non-trivial symmetry, providing a useful framework for comparing equivalent theories.

7 Conclusion

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A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned}
\delta S &= \delta \int \mathcal{L} dx^\mu \\
&= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho}\delta\eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho}\delta\partial_\nu\eta_\rho dx^\mu \\
&= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho}\delta\eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho}\delta\eta_\rho + \underbrace{\partial_\nu\left(\frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho}\delta\eta_\rho\right)}_{\text{border terme equal to zero}} dx^\mu \\
&= 0
\end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \tag{46}$$

which is in fact the equation of mouvement for the field